

表 6: 不同模型在多个数据集上的异常检测结果

模型	FlowGraph							TraceLog							ADFA-LD					HDFS								
	AUROC	AP	P@1%	R@1%	TPR@1%FPR	F1	MCC	AUROC	AP	P@1%	R@1%	TPR@1%FPR	F1	MCC	AUROC	AP	P@1%	R@1%	TPR@1%FPR	F1	MCC	AUROC	AP	P@1%	R@1%	TPR@1%FPR	F1	MCC
DeepTraLog	0.9785 ± 0.0228	0.9856 ± 0.0135	1.0000 ± 0.0000	0.0300 ± 0.0000	0.9667 ± 0.0289	0.9796 ± 0.0136	0.9644 ± 0.0235	0.6472 ± 0.0163	0.5416 ± 0.0430	0.6667 ± 0.5774	0.0193 ± 0.0167	0.0694 ± 0.0673	0.1228 ± 0.1171	0.1210 ± 0.1666	–	–	–	–	–	–	–	–	–	–	–	–	–	–
DeepTraLog-adapted	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0.7608 ± 0.0038	0.4394 ± 0.0061	0.6667 ± 0.1000	0.0268 ± 0.0040	0.0393 ± 0.0015	0.2270 ± 0.0412	0.1038 ± 0.0425	0.7379 ± 0.0120	0.6389 ± 0.0035	1.0000 ± 0.0000	0.0607 ± 0.0000	0.4941 ± 0.0018	0.6440 ± 0.0035	0.6349 ± 0.0051
GLocalKD	0.9561 ± 0.0082	0.9738 ± 0.0025	1.0000 ± 0.0000	0.0300 ± 0.0000	0.9500 ± 0.0000	0.9727 ± 0.0029	0.9526 ± 0.0054	0.6795 ± 0.0060	0.6318 ± 0.0057	1.0000 ± 0.0000	0.0289 ± 0.0000	0.2912 ± 0.0089	0.4388 ± 0.0152	0.4089 ± 0.0278	–	–	–	–	–	–	–	–	–	–	–	–	–	–
GLocalKD-adapted	–	–	–	–	–	–	–	–	–	–	–	–	–	–	0.7236 ± 0.0111	0.4164 ± 0.0134	0.6556 ± 0.3372	0.0264 ± 0.0136	0.0290 ± 0.0182	0.2503 ± 0.0229	0.1635 ± 0.0155	0.7559 ± 0.0057	0.6467 ± 0.0019	1.0000 ± 0.0000	0.0607 ± 0.0000	0.4963 ± 0.0018	0.6450 ± 0.0008	0.6348 ± 0.0000
HGT	0.5374 ± 0.4560	0.4960 ± 0.1967	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	−0.0693 ± 0.0745	0.6583 ± 0.0582	0.5532 ± 0.0662	0.5833 ± 0.3643	0.0169 ± 0.0105	0.1340 ± 0.1127	0.2277 ± 0.1195	0.2557 ± 0.1014	–	–	–	–	–	–	–	–	–	–	–	–	–	–
HRA-GNN	0.6646 ± 0.5489	0.7462 ± 0.4131	0.6667 ± 0.5774	0.0200 ± 0.0173	0.6333 ± 0.5485	0.6463 ± 0.5597	0.6110 ± 0.5810	0.8150 ± 0.0123	0.7497 ± 0.0225	1.0000 ± 0.0000	0.0289 ± 0.0000	0.2969 ± 0.0675	0.4482 ± 0.0790	0.4341 ± 0.0663	–	–	–	–	–	–	–	–	–	–	–	–	–	–
HRGCN	0.9333 ± 0.1155	0.8672 ± 0.2300	0.6667 ± 0.5774	0.0200 ± 0.0173	0.6667 ± 0.5774	0.6650 ± 0.5759	0.6438 ± 0.6093	0.7381 ± 0.0441	0.6961 ± 0.0321	1.0000 ± 0.0000	0.0289 ± 0.0000	0.3365 ± 0.0051	0.4795 ± 0.0318	0.4658 ± 0.0277	–	–	–	–	–	–	–	–	–	–	–	–	–	–
OCHetGCN	0.9557 ± 0.0561	0.8898 ± 0.1494	0.6667 ± 0.5774	0.0200 ± 0.0173	0.4200 ± 0.5188	0.5299 ± 0.5007	0.4754 ± 0.5257	0.7062 ± 0.0603	0.6015 ± 0.0551	0.6917 ± 0.2898	0.0200 ± 0.0084	0.1015 ± 0.1302	0.2167 ± 0.1662	0.2296 ± 0.1674	0.7938 ± 0.0030	0.5092 ± 0.0029	0.1444 ± 0.0509	0.0058 ± 0.0020	0.0027 ± 0.0000	0.5154 ± 0.0117	0.3973 ± 0.0108	0.7368 ± 0.0124	0.6380 ± 0.0101	1.0000 ± 0.0000	0.0607 ± 0.0000	0.4973 ± 0.0018	0.6467 ± 0.0051	0.6411 ± 0.0111